

PATH 1017

Seizures

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Seizures

- Abnormal, uncontrolled electrical discharge (firing) from a group of neurons in the cerebral cortex
- Leads to an abrupt and temporary alteration in cerebral activity
- Considered a symptom of underlying CNS dysfunction
- S+S vary according to site of discharge in brain

Theories on Causes of Seizures

- Neuron membrane alterations
- Altered ion distribution
- Decreased inhibition neuron activity
- Structural changes
- Neurotransmitter imbalances

Seizures in Canada

- 1 in 100 Canadians have epilepsy
- First seizure typically before 20 years of age
- First seizure after 20 years of age typically due to structural changes, trauma, tumour, stroke
- May occur at any age

Broad Categories

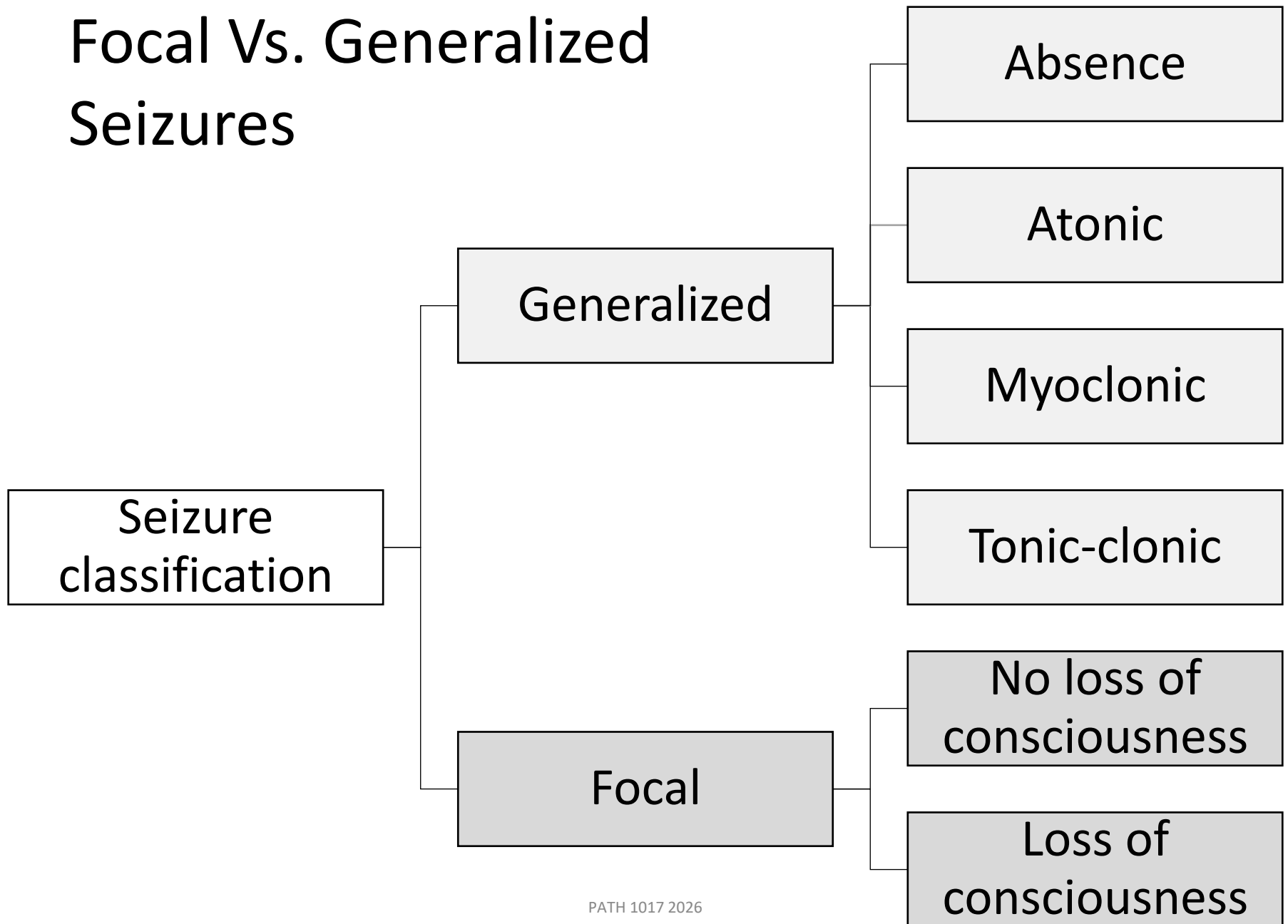
Unprovoked seizures:

- cause unknown
- Epilepsy

Provoked seizures

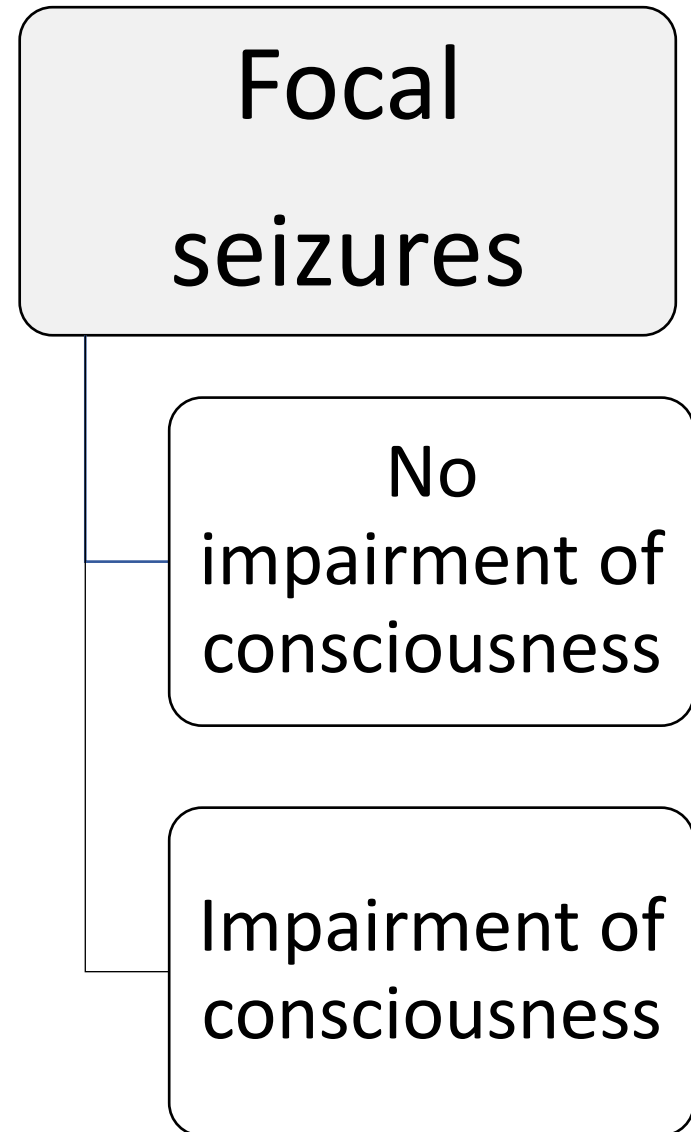
- secondary to the underlying cause
- Example: HI, BT, CNS infections
- ...

Focal Vs. Generalized Seizures



Focal Seizures

- One hemisphere involved
- Most common form of seizures in the setting of epilepsy
- Sub-classified depending of effect on consciousness or awareness



Focal Seizures Without Impairment Of Consciousness OR Awareness

- One hemisphere involved
 - may be localized or widespread within hemisphere
- No loss of consciousness or responsiveness
- Motor, sensory, autonomic symptoms
- Aura

What is an aura?

Focal Seizure With Impairment of Consciousness Or Awareness

- Begins in one hemisphere and rapidly spreads to the other hemisphere
 - Typically arise on temporal lobe
- Impairment of consciousness and awareness
 - May have hallucinations and illusional experiences
- Automatism
- Confusion post-ictually

What are automatisms?

What is a post-ictal state?

Generalized Seizures

- Both hemispheres involved at onset
- Unconsciousness
- Motor activity is symmetrical on both sides
- 6 broad categories exist

Types of Generalized Seizures

Absence

- disturbances in consciousness

Atonic

- loss of muscle tone

Myoclonic

- muscles strongly contract and jerk

Tonic-clonic

- muscle contraction & relaxation with loss of consciousness

Absence Seizures

- Disturbance in consciousness (**no convulsions**)
- Typically children only
- Manifestations:
 - Blank stare, motionless, unresponsive
 - Motions (automatisms)
 - Brief loss of contact with environment
 - Lasts a few seconds, then resumes normal activities immediately

Atonic Seizures

- Sudden loss of muscle tone
- Manifestations:
 - Slackening of jaw
 - Drooping of limbs
 - Falling to ground
- Risk for injury!!!

Myoclonic Seizure

- Brief involuntary muscle contractions
- Triggered by cerebral stimuli
- Manifestations:
 - Bilateral jerking of muscles
 - Generalized or confined to face, trunk, or one or more extremities
- Subtypes exist

Tonic-Clonic Seizures

- Major motor seizure
- May have a vague warning (focal seizure)
- Tonic phase: Sharp tonic contraction of muscles with extension of extremities
 - Immediate loss of consciousness
 - Bladder/bowel incontinence may occur
 - Cyanosis
- Clonic phase: Rhythmic bilateral contraction and relaxation of extremities
 - Unconscious until post-ictal phase –why?

Diagnostic Tests

- Neurological exam
- EEG
- Others:
 - CT or MRI
 - Labs to exclude other causes

Treatment

How do anti-epileptic drugs control seizure activity?

- Protect from injury during seizure
- Reduce seizure activity:
 - Pharmacological:
 - Anti-epileptic drugs dependent on seizure type
 - Examples: Phenytoin, Valproic acid
 - Lorazepam during seizure activity
 - Surgery may be considered if not responsive to pharmacological treatment with 2 or more drugs
- Neurostimulators

Complication: Status Epilepticus

- Seizures that do not stop spontaneously OR that occur in succession without recovery
- Clinical or electrical seizures lasting 30 minutes or more
- Usually tonic-clonic type seizures
- A **medical emergency** Leads to respiratory failure and death if no treatment
 - Requires life-support measures (airway!)
 - Medications to stop seizure and control
 - Identify underlying cause (i.e. tumour)

Prognosis

- With medications treatment:
 - Seizures eliminated in 1/3
 - Frequency reduce for 50%
 - 60% of pts who remain seizure-free can eventually stop their drugs and remain seizure-free
- Sudden unexplained death in epilepsy (SUDEP) is a rare complication , cause unknown